David W Miller

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Pipeline Software Engineer

Pipeline engineer and Technical Director with 10+ years building production-grade tools for AAA game and real-time content, with deep hands-on work inside Autodesk Maya alongside Python/C++ tooling across Unreal Engine and Houdini. Experience spans 15+ shipped productions diagnosing pipeline breakages, building utilities, and working with distributed art, engineering, and production teams to translate studio requirements into working software. Comfortable communicating complex technical tradeoffs to non-engineering stakeholders, working independently in a remote environment, and using AI-assisted coding tools to accelerate development.

TECHNICAL SKILLS

- Autodesk Maya: MEL scripting, Python API, M.A.S.H, Bifrost, Rigging, Blendshapes, mGear, nCloth, XGen. Production rigging, character setup and artist tool development
- Languages & Tooling: Python (PyQt/PySide, PIL, FFMpeg, REST/CRUD APIs), C++ (Unreal Editor Tooling), Perforce Helix Core, Git/GitHub Actions, Docker, Jenkins, MongoDB
- Pipeline & Production Systems: Flow Production Tracking, SyncSketch, AWS Deadline, Jira, Confluence
- Unreal Engine: UDK, 4.x, 5.x Editor & Real-time Blueprints, C++, Python, Plugin Development
- AI-Assisted Development: Claude and Codex for scripting, debugging, and prototyping of internal tools
- Cross-Discipline Collaboration: Partnered with animators, layout artists and engineers to develop live production fixes

EXPERIENCE

Technical Director | Halon Entertainment | Mar 2021 - Mar 2026

Building pipeline tools, lighting rigs, shader solutions and production automation systems supporting 20+ Unreal Engine and Maya projects spanning film, commercials and game cinematics including Fortnite, New World, Borderlands 3, Tiny Tina's Wonderlands, Remnant 2 and The Outer Worlds. Defined technical direction for distributed art teams, architected scalable toolsets, and provided hands-on engineering support across character, environment, and sequencer-based production.

- Engineered a Maya-to-Unreal publishing pipeline by extending AYON's open-source Unreal plugin in-house, automating asset and metadata synchronization between the two DCCs.
- Authored Python toolset to standardize Level Sequencer organization across all seasonal cinematic productions, conforming play ranges, synchronizing subsequences, and locking tracks to enforce pipeline consistency across large distributed teams.
- Developed gestalt character assembly pipeline for cinematic productions parsing game asset part lists and constructing complete cinematic-ready characters that could be directly handed off to animators and riggers.
- Built a turntable renderer system for Remnant 2's gestalt inventory system, allowing layout artists to visually scan creatures and environment assets to inform scene composition decisions.
- Implemented New Relic and Sentry production monitoring and observability stack for Maya and Unreal, establishing performance baselines, error tracking, and alerting workflows that reduced incident response time and improved system reliability.
- Provided hands-on Perforce Helix Core and UnrealGameSync (UGS) support, resolving workspace conflicts and sync failures to keep artists unblocked during live production.
- Built utility scripts to detect and resolve broken physics assets on characters within complex Sequencer hierarchies, reducing artist debugging time significantly across multi-season productions.
- Developed shaders for complex materials including glass, volumetric fog, and stylized water surfaces for real-time cinematic rendering in Unreal Engine.
- Delivered shader development, utility scripting, and artist support across all live productions, working directly within shipped game asset libraries.

Technical Artist | Halon Entertainment | Oct 2017 - Mar 2021

- Lead Technical Artist on previsualization for Call of the Wild and other films managing skinning, rigging, establishing art workflows and providing technical support for environment, animation and lighting artists.
- Provided technical art support for numerous virtual production shoots, troubleshooting real-time rendering issues, camera tracking integration, and LED wall calibration during live capture sessions.
- Optimized Unreal Engine environments for Intel and AMD graphics card demos, implementing LOD strategies, reducing material complexity and draw calls, and tuning lighting and post-processing to hit performance targets for retail showcase deployment.
- Collaborated with asset and character artists, animators, designers, and engineers to optimize real-time assets, integrate complex scenes, and build custom materials while maintaining high visual fidelity under production deadlines.

Solo Technical Developer | Winds of Almerra | 2017 - Present

- Engineered a C++ NPC AI system with a full behavior state machine and dynamic look-at/aim calculator, supporting context-driven transitions between melee/ranged combat, mounts, and siege weapons based on inventory, ammo, and health state.
- Built a headless Maya-to-Unreal armor pipeline to handle body deformations across all character skins, automating rigging, material assignment, and import for in-game items.
- Developed a headless Bifrost environment scatter system in Maya for procedural placement of rocks, trees, and rubble, importing directly into Unreal without manual intervention.
- Built a Landscape API integration to auto-import Houdini flowmaps into Unreal while preserving in-editor landscape painting, bridging a DCC-to-runtime terrain workflow.

Engine Artist & Game Developer | MatterVR | Feb 2017 - Aug 2017

- Shipped mobile VR game Wonders of the World for GearVR using early Unreal Engine 4, handling full technical art pipeline from asset integration to performance optimization.
- Built custom VR rendering pipeline, profiling and optimizing materials, meshes, and lighting to maintain consistent 60fps targets across mobile hardware constraints.
- Developed real-time multiplayer minigames in BP & C++, implementing networked gameplay for historical board games with data-driven progression systems.
- Created modular material systems and reusable art components, establishing technical standards that improved asset creation efficiency across the project.

JSC Distance Learning Instructional Designer | NASA JSC, GHG Corporation | July 2014 - Oct 2016

- Delivered +200 interactive online lessons to the Flight Operations Directorate software familiarization and classroom-based instruction, providing refresher briefings and training courses to flight controllers, instructors and astronauts.
- Developed customized training overviews for ISS laptop software by meeting with stakeholders and developers and hosting planning sessions.
- SharePoint developer & administrator, developing C# components supporting instructors and division pages across the Training Standards & Integration Production Group.
- Prototyped a fully-functional ISS Environment in Unreal Engine 4 for Interactive Emergency Training.

JSC Distance Learning Training Standards Intern | NASA JSC, United Space Alliance | Sep 2011 - Jul 2014

- Built 3 ISS Emergency Response online training courses using Articulate Storyline for Fire, Depress or Toxic Atmosphere scenarios using training footage from station mockups.
- Camera Operator for Video Lab capture of Flight Controller and Astronaut training footage and briefings.

EDUCATION

M.A. Digital Media Studies / University of Houston | Dec 2016
Compositing, Illustration, 3D Modelling & Animation

B.A. Communications / University of Houston | May 2011
TV Production Studio, School Newspaper & Magazine